

LEVEL 1 — TEST 2: BASELINE REPEAT

SEWAC Multi-Vehicle Telemetry Architecture

Abstract

This report presents the results of Level 1, Test 2, conducted as a repeat baseline evaluation of the SEWAC multi-vehicle telemetry architecture. Three vehicles operated concurrently for 10 minutes with randomized telemetry transmission intervals of 5–12 seconds. A total of 205 packets were generated and all 205 were reported as successfully transmitted, resulting in a 100% transmission success rate. The receiving-side observations recorded 105 packets across the three vehicle streams. This corresponds to an observed receiving rate of 51.22% and an unobserved packet shortfall of 100 packets, or 48.78%. The result confirms that the transmission stage remained stable while the receiving/buffering path continued to exhibit a substantial packet-processing bottleneck.

Index Terms

SEWAC, telemetry, multi-vehicle simulation, baseline testing, stress testing, buffering, receiving pipeline, IoT telemetry.

I. Introduction

Level 1, Test 2 was performed as a repeat baseline experiment following the initial baseline assessment. The purpose of the repeat test was to determine whether the behavior observed in the first baseline test persisted under substantially similar operating conditions.

Three vehicles transmitted telemetry concurrently over a ten-minute period. Packet intervals were randomized between 5 and 12 seconds, preventing the three streams from following a synchronized transmission schedule. The experiment therefore continued to represent independent concurrent vehicle traffic rather than a fixed periodic workload.

II. Test Configuration

Vehicle	Assigned Ward	Telemetry Unit
KA05AB1234	Ward 1	SEWAC_01_UHF
KA05AB1235	Ward 2	SEWAC_01_UHF
KA05AB1236	Ward 20	SEWAC_01_UHF

The test duration was 10 minutes. Each vehicle independently generated packets using a randomized delay between 5 and 12 seconds. No intentional transmission failures were introduced.

III. Test Objective

The objective was to repeat the Level 1 baseline workload and evaluate the consistency of the telemetry architecture. The test specifically examined whether the transmission layer continued to accept all generated packets and whether the receiving/buffering path could retain and process the transmitted traffic without a significant shortfall.

IV. Experimental Procedure

All three vehicle simulations were initiated concurrently. Each simulator generated telemetry independently and transmitted packets at randomized intervals. The transmission-side simulator recorded the success or failure of each submission. Receiving-side observations were then used to determine the number of packets actually observed for each vehicle.

The repeat-baseline configuration was intentionally kept at the same workload class as Test 1 so that changes in receiving behavior could be evaluated against the previously established baseline.

V. Results

A. Overall Transmission Results

Metric	Result
Test Level	Level 1
Test Number	2
Test Name	BASELINE
Duration	10 minutes
Random Packet Interval	5–12 seconds
Total Packets Sent	205
Successful Transmissions	205
Transmission Failures	0
Transmission Success Rate	100%

All 205 packets were successfully transmitted. The sending-side result therefore remained perfect, with no transmission failures recorded during the ten-minute test.

B. Receiving-End Results

Vehicle	Ward	Packets Actually Received
KA05AB1234	Ward 1	28
KA05AB1235	Ward 2	38
KA05AB1236	Ward 20	39
Total	—	105

The receiving-side observations recorded 105 packets in total. Compared with the 205 packets successfully transmitted, 100 packets were not observed at the receiving end.

C. Quantitative Performance Analysis

The observed receiving rate was calculated as $105/205 \times 100 = 51.22\%$. The corresponding receiving-side shortfall was $205 - 105 = 100$ packets, representing $100/205 \times 100 = 48.78\%$ of the transmitted packet volume.

VI. Vehicle-Level Analysis

Vehicle	Ward	Packets Received	Share of Received Packets
KA05AB1234	1	28	26.67%
KA05AB1235	2	38	36.19%
KA05AB1236	20	39	37.14%
Total	—	105	100%

The receiving observations were distributed across all three vehicles. KA05AB1236 recorded the highest number of observed packets with 39, followed by KA05AB1235 with 38 and KA05AB1234 with 28. The distribution demonstrates that the receiving path was active for all vehicle streams, although a substantial number of transmitted packets remained unobserved.

VII. Discussion

Test 2 confirms the stability of the transmission stage under the Level 1 workload. The system accepted all 205 generated packets without a single transmission failure. This is consistent with the behavior expected from a functioning packet-generation and initial-ingestion mechanism.

The receiving-side result improved relative to the previously recorded baseline observation, with 105 packets observed compared with 63 in the earlier test. The observed receiving rate therefore improved from 31.5% to 51.22%. Despite this improvement, the receiving stage still did not achieve complete packet retention or observation.

This result indicates that the principal architectural concern remains the receiving/buffering/processing path rather than the vehicle simulation or transmission mechanism. The repeat test also demonstrates why the receiving path must be instrumented independently: a 100% transmission success rate can coexist with a substantially lower downstream observation rate.

VIII. Bottleneck Assessment

The receiving-side shortfall is consistent with the presence of capacity limitations or bottlenecks within the downstream telemetry pipeline. Based on the observed behavior, the receiving and buffering stages should be treated as the critical performance boundary for subsequent testing.

The current test data does not establish the exact location of each dropped or unobserved packet. Therefore, the 100-packet shortfall should be classified as an observed receiving-side discrepancy rather than definitively attributed to a particular queue, consumer, database operation, or processing component.

IX. Comparison With Level 1, Test 1

Metric	Test 1	Test 2
Packets Sent	200	205
Transmission Success	200 / 200 (100%)	205 / 205 (100%)
Packets Received	63	105
Packets Not Observed	137	100
Observed Receiving Rate	31.5%	51.22%
Observed Shortfall	68.5%	48.78%

Compared with Test 1, Test 2 showed an improvement in the number of packets observed at the receiving end, increasing from 63 to 105 packets. The observed receiving rate therefore improved from 31.5% to 51.22%. Despite this improvement, the receiving stage still did not achieve complete packet retention or observation.

X. Limitations

The receiving-side counts supplied for this test represent packets actually observed and do not identify the precise point at which the remaining 100 packets became unavailable. No conclusion is therefore made regarding whether individual packets were dropped in transport, lost in a buffer, delayed beyond the observation window, rejected by a consumer, or affected by downstream processing constraints.

Additional instrumentation is required to distinguish packet arrival, queue insertion, queue removal, processing completion, database persistence, and final observation.

XI. Conclusion

Level 1, Test 2 successfully demonstrated that the SEWAC telemetry transmission stage remained stable under the baseline workload. All 205 generated packets were transmitted successfully, producing a 100% transmission success rate with zero reported failures.

At the receiving end, 105 packets were observed, producing a receiving rate of 51.22% and a shortfall of 100 packets. Although this represents a substantial improvement over the receiving result recorded in Test 1, the receiving/buffering pipeline continues to demonstrate a significant bottleneck.

Accordingly, Level 1, Test 2 is classified as a PARTIAL PASS. The transmission layer passes the baseline requirement, while the receiving and buffering path remains the primary area requiring optimization and further instrumentation before higher stress levels are introduced.

XII. Recommended Next Action

Before advancing to higher stress levels, the telemetry pipeline should record queue depth, packets enqueued, packets dequeued, consumer throughput, processing latency, rejected packets, and persisted records. These measurements will allow subsequent tests to distinguish between buffer saturation, consumer throughput limitations, processing delays, and downstream persistence issues.

Table I — Final Level 1 Test 2 Summary

Parameter	Observed Result
Test	Level 1 — Test 2
Scenario	Baseline Repeat
Duration	10 minutes
Vehicles	3
Packets Sent	205
Transmission Success	205 / 205 (100%)
Transmission Failures	0
Packets Received	105
Packets Not Observed	100
Observed Receiving Rate	51.22%
Observed Receiving Shortfall	48.78%
Overall Assessment	PARTIAL PASS